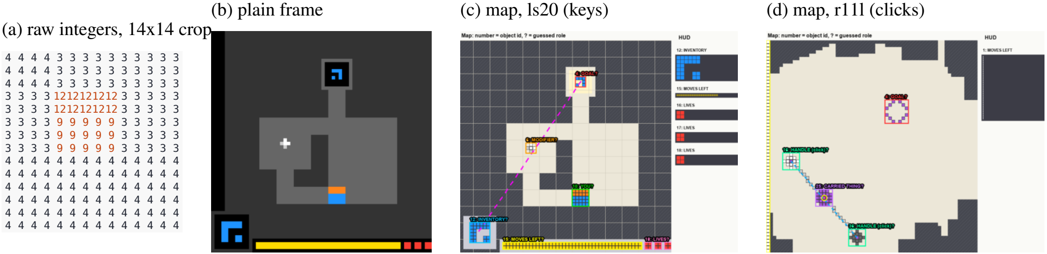

(a) raw integers, 14x14 crop

```
4 4 4 4 3 3 3 3 3 3 3 3 3 3
4 4 4 4 3 3 3 3 3 3 3 3 3 3
4 4 4 4 3 3 3 3 3 3 3 3 3 3
3 3 3 3 1212121212 3 3 3 3 3
3 3 3 3 1212121212 3 3 3 3 3
3 3 3 3 9 9 9 9 9 3 3 3 3 3
3 3 3 3 9 9 9 9 9 3 3 3 3 3
3 3 3 3 9 9 9 9 9 3 3 3 3 3
4 4 4 4 4 4 4 4 4 4 4 4 4 4
4 4 4 4 4 4 4 4 4 4 4 4 4 4
4 4 4 4 4 4 4 4 4 4 4 4 4 4
4 4 4 4 4 4 4 4 4 4 4 4 4 4
4 4 4 4 4 4 4 4 4 4 4 4 4 4
4 4 4 4 4 4 4 4 4 4 4 4 4 4
```

(b) plain frame

(c) map, ls20 (keys)

Map: number = object id, ? = guessed role
HUD
12: INVENTORY
15: MOVES LEFT
16: LIVES
17: LIVES
18: LIVES
4: GOAL?
6: MODIFIER?
10: YOU?
12: INVENTORY?
15: MOVES LEFT?
18: LIVES?

(d) map, r11l (clicks)

Map: number = object id, ? = guessed role
HUD
1: MOVES LEFT
4: GOAL?
16: HANDLE (click)?
25: CARRIED THING?
36: HANDLE (click)?